Lecture 21: Positive Definite Matrices

Quadratic Forms, Ellipses, and the Question $A = M^T M$

The Question

Today's Question

Last time, one matrix appeared again and again:

$$M^T M.$$

For every vector x:

$$x^T M^T M x = (Mx)^T (Mx) = \|Mx\|^2.$$

So $M^T M$ always produces a **squared length**.

Which symmetric matrices can be written as $M^T M$?

The Necessary Condition

If

$$A = M^T M,$$

then:

$$x^T A x = x^T M^T M x = \|Mx\|^2.$$

Therefore:

$$\boxed{x^T A x \geq 0 \quad \text{for every } x.}$$

So a candidate A must never assign negative squared length.

A Symmetric Matrix Measures Length?

A real symmetric matrix A gives a quadratic expression:

$$q_A(x) = x^T Ax.$$

If $q_A(x)$ behaves like squared length, then:

$$q_A(x) > 0 \quad \text{for every } x \neq 0.$$

If $q_A(x)$ sometimes becomes negative, it cannot be a squared length.

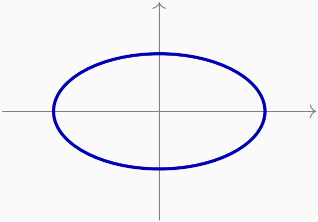
The whole lecture is about testing this sign.

Look at the Unit Shape

Instead of testing every vector one by one, draw the set:

$$\{x : x^T A x = 1\}.$$

This is the "unit circle" for the quadratic form.



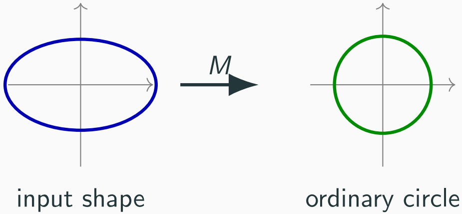
$x^T A x = 1$

If It Comes From $M^T M$

If $A = M^T M$, then:

$$x^T A x = 1 \iff \|Mx\|^2 = 1.$$

So the shape is a preimage of an ordinary circle/sphere.



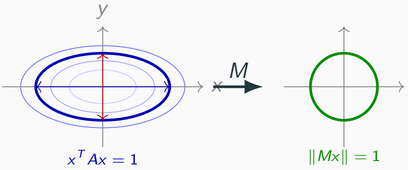
This suggests ellipse-type geometry, not hyperbola geometry.

Three Possible Shapes

Good Case: Ellipse

$$A = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & 1 \end{pmatrix}, \qquad M = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}, \qquad A = M^T M.$$

$$x^T A x = 1 \Longleftrightarrow \frac{x^2}{4} + y^2 = 1.$$



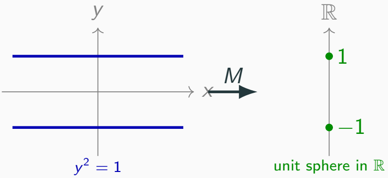
$$Mx = \begin{pmatrix} x/2 \\ y \end{pmatrix}, \qquad x^T A x = \|Mx\|^2.$$

After applying M, the ellipse becomes the unit circle.

Borderline Case: Two Lines

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad M = \begin{pmatrix} 0 & 1 \end{pmatrix}, \qquad A = M^T M.$$

$$x^T A x = 1 \iff y^2 = 1.$$



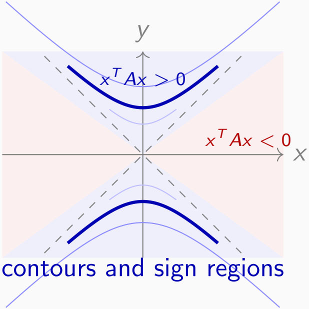
$M(x, y) = y$ projects onto the one-dimensional target $\mathbb{R}$.

The equation becomes $\|M(x, y)\|^2 = 1$, so $M(x, y) = \pm 1$.

Two points are the unit sphere in one-dimensional space.

Bad Case: Hyperbola

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad x^T A x = 1 \iff -x^2 + y^2 = 1.$$



contours and sign regions

There is no real matrix M with $A = M^T M$: squared lengths cannot become negative.

Three Shapes, Three Behaviors

Matrix	**Shape of $x^T A x = 1$**	**Behavior**
$\begin{pmatrix} \frac{1}{4} & 0 \\ 0 & 1 \end{pmatrix}$	ellipse	always positive
$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$	two lines	never negative, but degenerate
$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	hyperbola	sometimes negative

Now we need names for these three behaviors.

Definitions

Definition

Let A be a real symmetric matrix.

- A is **positive definite** if
$$x^T Ax > 0 \quad \text{for every } x \neq 0.$$
- A is **positive semidefinite** if
$$x^T Ax \geq 0 \quad \text{for every } x.$$
- A is **negative definite** if
$$x^T Ax < 0 \quad \text{for every } x \neq 0.$$
- A is **negative semidefinite** if
$$x^T Ax \leq 0 \quad \text{for every } x.$$
- A is **indefinite** if $x^T Ax$ takes both positive and negative values.

Definite Means Semidefinite

Positive definite is a stronger condition:

$$x^T A x > 0 \text{ for every } x \neq 0 \quad \Longrightarrow \quad x^T A x \geq 0 \text{ for every } x.$$

So:

$$\boxed{\text{positive definite} \quad \Longrightarrow \quad \text{positive semidefinite.}}$$

Similarly:

$$\boxed{\text{negative definite} \quad \Longrightarrow \quad \text{negative semidefinite.}}$$

Semidefinite is the broader category; definite is the strict case.

Entry Point: Which Matrices Are $M^T M$?

Theorem (Factorization Criterion)

For a real symmetric matrix A,

$$\boxed{A = M^T M \text{ for some real matrix } M} \quad \Longleftrightarrow \quad \boxed{A \text{ is positive semidefinite.}}$$

One direction is immediate:

$$A = M^T M \quad \Longrightarrow \quad x^T A x = x^T M^T M x = \|Mx\|^2 \geq 0.$$

For positive definite, the strengthened statement is:

$$\boxed{A \text{ is positive definite} \quad \Longleftrightarrow \quad A = M^T M \text{ for some full-column-rank } M.}$$

Indeed, if M has full column rank, then $x \neq 0$ implies $Mx \neq 0$, hence

$$x^T M^T M x = \|Mx\|^2 > 0.$$

The converse will follow once eigenvalues give square roots.

Workable Theorem Preview

The computational theorem we will prove is:

Theorem (Diagonal Cross-Filling Test)

Perform diagonal cross-filling on a real symmetric matrix A.

- If the process gets stuck at a zero diagonal with a nonzero row/column, then A is **indefinite**.
- If the nonzero pivots contain both positive and negative numbers, then A is **indefinite**.
- If all pivots are nonnegative and diagonal cross-filling proceeds with no difficulties, then A is **positive semidefinite**.
- If all pivots are nonpositive and diagonal cross-filling proceeds with no difficulties, then A is **negative semidefinite**.
- If the number of positive pivots equals the size of the matrix, then A is **positive definite**.
- If the number of negative pivots equals the size of the matrix, then A is **negative definite**.

This is the course-native test; eigenvalues are a second language.

Matrix as an Inner Product Table

Where Does $x^T A x$ Come From?

The ordinary dot product is:

$$\langle v, w \rangle = v^T w.$$

A symmetric matrix A gives a new bilinear form:

$$\boxed{\langle v, w \rangle_A = v^T A w.}$$

Then the quadratic form is the self-pairing:

$$\boxed{\langle x, x \rangle_A = x^T A x.}$$

So A is trying to define a new geometry.

Symmetry Means $A = A^T$

For an inner-product-like geometry, we want:

$$\langle v, w \rangle_A = \langle w, v \rangle_A.$$

Compute:

$$\langle v, w \rangle_A = v^T A w.$$

Also:

$$\langle w, v \rangle_A = w^T A v = (w^T A v)^T = v^T A^T w.$$

So symmetry of the form is exactly:

$$\boxed{A = A^T.}$$

Definition: Standard Basis Vectors

In $\mathbb{R}^3$:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

The vector e_i is the coordinate probe:

$$e_i^T A e_j = \text{the } (i, j) \text{ entry of } A.$$

$$e_2^T A e_3 = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = a_{23}.$$

Matrix Entries Are Inner Products

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} \langle e_1, e_1 \rangle_A & \langle e_1, e_2 \rangle_A & \langle e_1, e_3 \rangle_A \\ \langle e_2, e_1 \rangle_A & \langle e_2, e_2 \rangle_A & \langle e_2, e_3 \rangle_A \\ \langle e_3, e_1 \rangle_A & \langle e_3, e_2 \rangle_A & \langle e_3, e_3 \rangle_A \end{pmatrix}}_{\text{inner product table of basis vectors}}.$$

$$\boxed{\underbrace{a_{ij}}_{\text{matrix entry}} = \underbrace{e_i^T A e_j}_{\langle e_i, e_j \rangle_A}.}$$

The matrix A is the inner product table of the standard basis.

Standard Basis Vectors

In $\mathbb{R}^3$:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

The vector e_2 means:

select the second coordinate.

So multiplying by e_2 should select the second column or second row.

Right Multiplication Selects a Column

Let

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Then:

$$Ae_2 = \begin{pmatrix} a_{12} \\ a_{22} \\ a_{32} \end{pmatrix}.$$

Ae_2 selects column 2.

Left Multiplication Selects a Row

Now put e_2^T on the left:

$$e_2^T = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}.$$

Then:

$$e_2^T A = \begin{pmatrix} a_{21} & a_{22} & a_{23} \end{pmatrix}.$$

$e_2^T A$ selects row 2.

Diagonal Entry = Pivot Value

Put both together:

$$e_2^T A e_2 = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Result:

$$\boxed{e_2^T A e_2 = a_{22}.}$$

This is the diagonal pivot used later in diagonal cross-filling.

Basis Table Picture

The whole matrix is a table:

$$A = \begin{pmatrix} \langle e_1, e_1 \rangle_A & \langle e_1, e_2 \rangle_A & \langle e_1, e_3 \rangle_A \\ \langle e_2, e_1 \rangle_A & \langle e_2, e_2 \rangle_A & \langle e_2, e_3 \rangle_A \\ \langle e_3, e_1 \rangle_A & \langle e_3, e_2 \rangle_A & \langle e_3, e_3 \rangle_A \end{pmatrix}.$$

The diagonal entries are self-lengths:

$$a_{ii} = e_i^T A e_i = \langle e_i, e_i \rangle_A.$$

Eigenvalue Criterion

A Fast Positive Example

Consider

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

Its eigenvectors are:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The eigenvalues are:

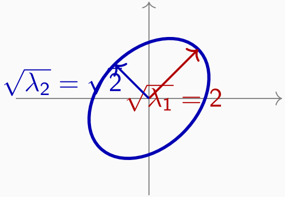
$$\lambda_1 = 4, \qquad \lambda_2 = 2.$$

Both eigenvalues are positive: ellipse geometry.

Ellipse from Positive Eigenvalues

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

$$\lambda_1 = 4, \quad \lambda_2 = 2.$$



$$\underbrace{\sqrt{\lambda_1} = 2}_{\text{semi-axis length}}, \qquad \underbrace{\sqrt{\lambda_2} = \sqrt{2}}_{\text{semi-axis length}}.$$

Eigenvalues are squared coefficients; semi-axis lengths use square roots.

A Fast Negative Example

Now consider

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

The same two directions appear:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

But the eigenvalues are:

$$\lambda_1 = 3, \qquad \lambda_2 = -1.$$

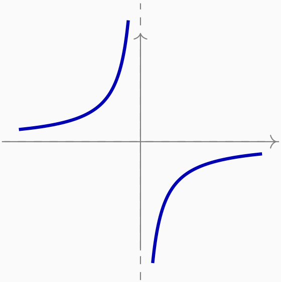
One negative eigenvalue: not squared length.

Hyperbola from a Negative Eigenvalue

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$\lambda_1 = 3, \quad \lambda_2 = -1.$$

One direction contributes negative squared length.



A negative eigenvalue produces hyperbola-type geometry.

Eigenvalue Criterion

Theorem (Eigenvalue Criterion)

Let A be a real symmetric matrix.

- A is positive definite iff all eigenvalues are positive.
- A is positive semidefinite iff all eigenvalues are nonnegative.
- A is indefinite if it has both positive and negative eigenvalues.

This is the fastest conceptual test.

Why This Criterion Works

From the real symmetric spectral theorem:

$$A = \Omega \Lambda \Omega^T, \qquad \Omega^T \Omega = I.$$

Then:

$$x^T A x = x^T \Omega \Lambda \Omega^T x.$$

Set:

$$y = \Omega^T x.$$

So:

$$x^T A x = y^T \Lambda y.$$

Diagonal Form Shows the Signs

If

$$\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \end{pmatrix}, \qquad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \end{pmatrix},$$

then:

$$y^T \Lambda y = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots .$$

The signs of the eigenvalues control the signs of $x^T A x$.

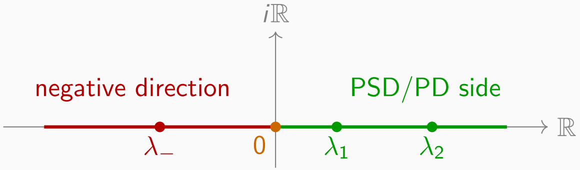
Eigenvalue Criterion Summary

Eigenvalues	Form	Shape
all > 0	positive except 0	ellipse
all ≥ 0	never negative	degenerate
mixed signs	both signs	hyperbola

The spectral theorem turns positivity into a diagonal sign check.

Eigenvalue Picture: Continue Normal Matrices

For real symmetric matrices, all eigenvalues lie on the real line.



This continues the normal-matrix eigenvalue picture: signs now decide geometry.

Close the Factorization Criterion

Suppose A is positive semidefinite.

By the eigenvalue criterion:

$$A = \Omega \Lambda \Omega^T, \qquad \lambda_i \geq 0.$$

Take square roots:

$$\sqrt{\Lambda} = \mathrm{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots).$$

Then:

$$A = \Omega \sqrt{\Lambda} \sqrt{\Lambda} \Omega^T = (\sqrt{\Lambda} \Omega^T)^T (\sqrt{\Lambda} \Omega^T).$$

So $M = \sqrt{\Lambda} \Omega^T$ proves $A = M^T M$.

$M^T M$: Semidefinite vs. Definite

The factorization alone gives only:

$$x^T M^T M x = \|Mx\|^2 \geq 0.$$

To get **positive definite**, we also need:

$$x \neq 0 \quad \Longrightarrow \quad Mx \neq 0.$$

Equivalently:

$$\boxed{\mathrm{Null}(M) = 0} \qquad \Longleftrightarrow \qquad \boxed{\text{columns of } M \text{ are linearly independent.}}$$

$M^T M$ is always PSD; it is PD exactly when M has full column rank.

Checking Semipositivity by Diagonal Cross-Filling

Workable Theorem

Theorem (Diagonal Cross-Filling Test)

Perform diagonal cross-filling on a real symmetric matrix A.

1. If the process gets stuck at a zero diagonal with a nonzero row/column, then A is **indefinite**.
2. If the nonzero pivots contain both positive and negative numbers, then A is **indefinite**.
3. If all pivots are nonnegative and diagonal cross-filling proceeds with no difficulties, then A is **positive semidefinite**.
4. If all pivots are nonpositive and diagonal cross-filling proceeds with no difficulties, then A is **negative semidefinite**.
5. If the number of positive pivots equals the size of the matrix, then A is **positive definite**.
6. If the number of negative pivots equals the size of the matrix, then A is **negative definite**.

$$\underbrace{\text{pivot signs}}_{\text{positive, zero, negative}} \qquad \underbrace{\text{zero row/column rule}}_{\text{no difficulty}} \qquad \underbrace{\text{contradictory signs}}_{\text{indefinite}}.$$

Running Examples Immediately

Matrix	pivots	no difficulties?	conclusion
$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$	$2,\ 3$	yes	positive definite
$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$	$2,\ 0$	yes: zero row/column	positive semidefinite
$\begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$	$2,\ -3$	yes	indefinite
$\begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}$	$-2,\ -3$	yes	negative definite

The theorem is a sign-reading rule for the pivots that cross-filling exposes.

Running Example: Stuck Means Indefinite

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}.$$

If we inspect the second diagonal first, we see:

$$a_{22} = 0, \qquad \text{but row/column 2 is not zero.}$$

Indeed:

$$\begin{pmatrix} t & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} t \\ 1 \end{pmatrix} = t^2 + 4t,$$

which is positive for large t and negative for small negative t.

A stuck zero pivot is not semidefinite; it reveals indefiniteness.

Proof Plan: Two Directions

$$\underbrace{\text{nonnegative pivots with no difficulties}}_{\substack{\text{positive pivots make rank-one pieces} \\ \text{zero rows/columns are skipped}}} \implies \underbrace{A = MM^T}_{\text{stack cross-filled columns}} \implies \underbrace{A \succeq 0}_{\text{easy direction}} .$$

$$\underbrace{A \succeq 0}_{\text{hard direction}} \implies \underbrace{\text{cross-filling proceeds with no difficulties}}_{\text{needs three modules}} .$$

> This proves the PSD half of the theorem.
> Apply the same logic to $-A$ for the NSD half.

Easy Direction: Positive Pivots Build Columns

$$\underbrace{\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}}_{\text{pivot column}} \qquad \underbrace{b_2 > 0}_{\text{positive pivot}} \qquad \Longrightarrow \qquad \underbrace{m = \frac{1}{\sqrt{b_2}} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}}_{\text{cross-filled column}}.$$

$$\underbrace{mm^T}_{\text{one PSD piece}} = \underbrace{\frac{1}{b_2} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \begin{pmatrix} b_1 & b_2 & b_3 \end{pmatrix}}_{\text{column times row}} = \underbrace{\begin{pmatrix} \frac{b_1^2}{b_2} & b_1 & \frac{b_1 b_3}{b_2} \\ b_1 & b_2 & b_3 \\ \frac{b_3 b_1}{b_2} & b_3 & \frac{b_3^2}{b_2} \end{pmatrix}}_{P}.$$

The positive pivot is required because the column uses $\sqrt{b_2}$.

Easy Direction: Stack All Cross-Filled Columns

$$\underbrace{A}_{\text{after successful cross-filling}} = \underbrace{m_1 m_1^T + m_2 m_2^T + \cdots + m_r m_r^T}_{\text{rank-one PSD pieces}}.$$

$$\underbrace{M = \begin{pmatrix} | & | & & | \\ m_1 & m_2 & \cdots & m_r \\ | & | & & | \end{pmatrix}}_{\text{stack the cross-filled columns}} \implies \underbrace{A = MM^T}_{\text{factorization}}.$$

$$\underbrace{x^T A x}_{\text{test vector}} = \underbrace{x^T M M^T x}_{\text{factorized form}} = \underbrace{\|M^T x\|^2}_{\geq 0}.$$

Hard Direction: Three Modules

$$\underbrace{A \succeq 0}_{\text{input assumption}} \quad \Longrightarrow \quad \begin{cases} \underbrace{a_{ii} = e_i^T A e_i \geq 0}_{\text{Module 1: no negative diagonal}}, \\ \underbrace{a_{ii} = 0 \Rightarrow \text{row } i \text{ and column } i \text{ are zero}}_{\text{Module 2: zero pivot is harmless}}, \\ \underbrace{a_{ii} > 0 \Rightarrow A = P + R, \quad P \succeq 0, \quad R \succeq 0}_{\text{Module 3: positive pivot keeps induction alive}}. \end{cases}$$

$$\boxed{\text{zero pivot} \;\Rightarrow\; \text{skip} \qquad \text{positive pivot} \;\Rightarrow\; \text{cross-fill and continue.}}$$

One Diagonal Cross-Fill Step

Use the second diagonal entry as the visible pivot:

$$a_{22} = e_2^T A e_2 > 0.$$

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} p_{11} & a_{12} & p_{13} \\ a_{21} & a_{22} & a_{23} \\ p_{31} & a_{32} & p_{33} \end{pmatrix}}_{P} + \underbrace{\begin{pmatrix} r_{11} & 0 & r_{13} \\ 0 & 0 & 0 \\ r_{31} & 0 & r_{33} \end{pmatrix}}_{R=A-P}.$$

Diagonal cross-filling means: honestly write the two matrices P and R.

Entries of the Rank-One Piece

At the pivot a_{22}, write the pivot column entries as

$$b_1 = a_{12}, \qquad b_2 = a_{22}, \qquad b_3 = a_{32}.$$

$$\underbrace{\begin{pmatrix} \frac{b_1 b_1}{b_2} & b_1 & \frac{b_1 b_3}{b_2} \\ b_1 & \color{red}{b_2} & b_3 \\ \frac{b_3 b_1}{b_2} & b_3 & \frac{b_3 b_3}{b_2} \end{pmatrix}}_{P}$$

This is the honest entrywise matrix display of the rank-one piece.

Which Entries Are Being Used?

row 2: $e_2^T A$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

column 2: Ae_2

pivot $a_{22} = e_2^T A e_2$

Cross-filling uses the pivot row and column:

$$p_{ij} = \frac{a_{i2} a_{2j}}{a_{22}}.$$

The formula is not abstract: it fills entries from the highlighted cross.

Inner Product Meaning of the Entries

$$\underbrace{\begin{pmatrix} \dfrac{\langle e_1, e_2\rangle_A \langle e_2, e_1\rangle_A}{\langle e_2, e_2\rangle_A} & \langle e_1, e_2\rangle_A & \dfrac{\langle e_1, e_2\rangle_A \langle e_2, e_3\rangle_A}{\langle e_2, e_2\rangle_A} \\ \langle e_2, e_1\rangle_A & \langle e_2, e_2\rangle_A & \langle e_2, e_3\rangle_A \\ \dfrac{\langle e_3, e_2\rangle_A \langle e_2, e_1\rangle_A}{\langle e_2, e_2\rangle_A} & \langle e_3, e_2\rangle_A & \dfrac{\langle e_3, e_2\rangle_A \langle e_2, e_3\rangle_A}{\langle e_2, e_2\rangle_A} \end{pmatrix}}_{P \text{ written entrywise from colored basis vectors}}$$

$$\underbrace{e_1}_{\text{first row/column}} \qquad \underbrace{e_2}_{\text{pivot direction}} \qquad \underbrace{e_3}_{\text{third row/column}} \quad .$$

Only the vectors are colored; each inner product remains one object.

Module 1: No Negative Diagonal

$$\underbrace{A \succeq 0}_{\text{PSD assumption}} \implies \underbrace{a_{ii} = e_i^T A e_i \geq 0}_{\text{test vector } e_i}.$$

$$\boxed{\underbrace{a_{ii} < 0}_{\text{negative diagonal appears}} \implies \underbrace{A \not\succeq 0}_{\text{stop cross-filling test}}.}$$

This explains why negative pivots immediately fail the PSD test.

Lemma: Cauchy Is the Tool for Continuation

$$\underbrace{A \succeq 0}_{\text{PSD form}} \quad \Longrightarrow \quad \underbrace{\langle x, y\rangle_A^2 \le \langle x, x\rangle_A \langle y, y\rangle_A}_{\text{Cauchy lemma}}.$$

Proposition (Cauchy Inequality for PSD Forms)

If A is positive semidefinite and $\langle x, y\rangle_A = x^T A y$, then

$$\langle x, y\rangle_A^2 \le \langle x, x\rangle_A \langle y, y\rangle_A.$$

$$\boxed{\begin{array}{lcl} \text{Use 1} & : & a_{ii} = 0 \Rightarrow \text{zero row/column}, \\ \text{Use 2} & : & a_{ii} > 0 \Rightarrow R = A - P \succeq 0. \end{array}}$$

The purpose of Cauchy is to keep cross-filling alive.

Module 2: Zero Diagonal Means Zero Row

Proposition (Zero Diagonal Lemma)

If $A \succeq 0$ and

$$a_{ii} = e_i^T A e_i = 0,$$

then the whole i-th row and the whole i-th column are zero.

For any j, Cauchy gives:

$$a_{ij}^2 = \langle e_i, e_j \rangle_A^2 \leq \langle e_i, e_i \rangle_A \langle e_j, e_j \rangle_A = 0 \cdot a_{jj} = 0.$$

So $a_{ij} = 0$ for every j.

Module 2 Meaning: Skip Zero Rows

If $A \succeq 0$ and a diagonal entry is zero:

$$a_{ii} = 0,$$

then the whole row and column are already zero.

$$\begin{pmatrix} * & * & 0 & * \\ * & * & 0 & * \\ 0 & 0 & 0 & 0 \\ * & * & 0 & * \end{pmatrix}$$

There is no need to choose this as a pivot. Skip it.

Module 3: Positive Pivot Splits PSD

Now assume the diagonal pivot is positive:

$$a_{ii} = e_i^T A e_i > 0.$$

Proposition (Positive Pivot Step)

If $A \succeq 0$ and $a_{ii} > 0$, then diagonal cross-filling gives

$$A = P + R, \qquad P \succeq 0, \qquad R \succeq 0.$$

Now prove it by looking at the actual entries of P and R.

Module 3 Setup: Write A as Inner Products

$$\underbrace{\langle e_2, e_2\rangle_A > 0}_{\text{positive pivot}}$$

$$\underbrace{\begin{pmatrix} \langle e_1, e_1\rangle_A & \langle e_1, e_2\rangle_A & \langle e_1, e_3\rangle_A \\ \langle e_2, e_1\rangle_A & \langle e_2, e_2\rangle_A & \langle e_2, e_3\rangle_A \\ \langle e_3, e_1\rangle_A & \langle e_3, e_2\rangle_A & \langle e_3, e_3\rangle_A \end{pmatrix}}_{A \text{ as an inner product table}}$$

Only the basis vectors are colored. Each inner product stays one object.

Module 3: The Cross-Filled Piece P

$$\underbrace{\begin{pmatrix} \frac{\langle e_1, e_2\rangle_A \langle e_2, e_1\rangle_A}{\langle e_2, e_2\rangle_A} & \langle e_1, e_2\rangle_A & \frac{\langle e_1, e_2\rangle_A \langle e_2, e_3\rangle_A}{\langle e_2, e_2\rangle_A} \\ \langle e_2, e_1\rangle_A & \langle e_2, e_2\rangle_A & \langle e_2, e_3\rangle_A \\ \frac{\langle e_3, e_2\rangle_A \langle e_2, e_1\rangle_A}{\langle e_2, e_2\rangle_A} & \langle e_3, e_2\rangle_A & \frac{\langle e_3, e_2\rangle_A \langle e_2, e_3\rangle_A}{\langle e_2, e_2\rangle_A} \end{pmatrix}}_{P}$$

$$\underbrace{P = mm^T}_{\text{PSD piece}}, \qquad \underbrace{m = \frac{1}{\sqrt{\langle e_2, e_2\rangle_A}} \begin{pmatrix} \langle e_1, e_2\rangle_A \\ \langle e_2, e_2\rangle_A \\ \langle e_3, e_2\rangle_A \end{pmatrix}}_{\text{positive pivot gives the square root}} .$$

What P Does to a Quadratic Form

The cross-filled piece is

$$P = mm^T, \qquad m = \frac{1}{\sqrt{\langle e_2, e_2\rangle_A}} \begin{pmatrix} \langle e_1, e_2\rangle_A \\ \langle e_2, e_2\rangle_A \\ \langle e_3, e_2\rangle_A \end{pmatrix}.$$

For

$$v = x_1 e_1 + x_2 e_2 + x_3 e_3,$$

the row-vector product with m is

$$\begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} m = \frac{\langle v, e_2\rangle_A}{\sqrt{\langle e_2, e_2\rangle_A}}.$$

P measures only the component of v seen by the pivot direction e_2.

Therefore $v^T P v$ Has One Square

Since $P = mm^T$,

$$
\begin{aligned}
v^T P v &= \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} mm^T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \\
&= \left(\begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} m \right)^2 \\
&= \boxed{\frac{\langle v, e_2 \rangle_A^2}{\langle e_2, e_2 \rangle_A}}.
\end{aligned}
$$

This is the formula used when testing the remainder $R = A - P$.

Module 3: The Remainder $R = A - P$

$$\underbrace{A-P}^{R} = \underbrace{\begin{pmatrix} \langle e_1, e_1\rangle_A - \dfrac{\langle e_1, e_2\rangle_A \langle e_2, e_1\rangle_A}{\langle e_2, e_2\rangle_A} & 0 & \langle e_1, e_3\rangle_A - \dfrac{\langle e_1, e_2\rangle_A \langle e_2, e_3\rangle_A}{\langle e_2, e_2\rangle_A} \\ 0 & 0 & 0 \\ \langle e_3, e_1\rangle_A - \dfrac{\langle e_3, e_2\rangle_A \langle e_2, e_1\rangle_A}{\langle e_2, e_2\rangle_A} & 0 & \langle e_3, e_3\rangle_A - \dfrac{\langle e_3, e_2\rangle_A \langle e_2, e_3\rangle_A}{\langle e_2, e_2\rangle_A} \end{pmatrix}}_{\text{entrywise remainder}}$$

The pivot row and pivot column are zero; only the remaining block must stay PSD.

Module 3: Test the Remainder

$$\underbrace{v}_{\text{testing vector}} = \underbrace{x_1 e_1 + x_2 e_2 + x_3 e_3}_{\text{any vector}}.$$

$$\underbrace{v^T R v}_{\text{test } R \text{ on both sides}} = \underbrace{v^T (A - P) v}_{R=A-P} = \underbrace{\langle v, v \rangle_A}_{v^T A v} - \underbrace{\frac{\langle v, e_2 \rangle_A^2}{\langle e_2, e_2 \rangle_A}}_{v^T P v}.$$

$$\underbrace{\langle v, e_2 \rangle_A^2}_{\text{Cauchy numerator}} \le \underbrace{\langle v, v \rangle_A \langle e_2, e_2 \rangle_A}_{\text{Cauchy denominator clears}} \implies \boxed{v^T R v \ge 0.}$$

Cauchy is exactly the reason the remainder remains positive semidefinite.

Cross-Filling Criterion So Far

For $A \succeq 0$:

Diagonal pivot	What happens
$a_{ii} < 0$	impossible
$a_{ii} = 0$	row/column must be zero; skip it
$a_{ii} > 0$	cross-fill: $A = P + (A - P)$ with both PSD

This is the diagonal cross-filling behavior we wanted.

Proof of Cauchy: Geometric Memory

Proposition (Cauchy Inequality for PSD Forms)

If $A \succeq 0$ and $\langle x, y \rangle_A = x^T A y$, then

$$\boxed{\langle x, y \rangle_A^2 \leq \langle x, x \rangle_A \langle y, y \rangle_A.}$$

$$\underbrace{A = I}_{\text{ordinary dot product}} \quad \Longrightarrow \quad \underbrace{(x^T y)^2 \leq (x^T x)(y^T y)}_{\text{circle intuition only}}.$$

This page is only for geometric memory, not the proof.

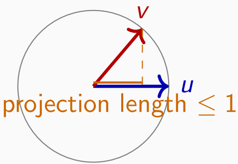
Cauchy: Picture

For the ordinary dot product:

$$(x^T y)^2 \leq (x^T x)(y^T y).$$

After normalizing:

$$\left(\frac{x}{\|x\|}\right)^T \left(\frac{y}{\|y\|}\right) \leq 1.$$



Cauchy Proof: Two Cases

Proposition (Cauchy Inequality for PSD Forms)

If $A \succeq 0$, then

$$\boxed{\langle x, y\rangle_A^2 \leq \langle x, x\rangle_A \langle y, y\rangle_A.}$$

Case	Proof method
$\langle x, x\rangle_A > 0$	build a cancellation vector w
$\langle x, x\rangle_A = 0$	use the polynomial in t

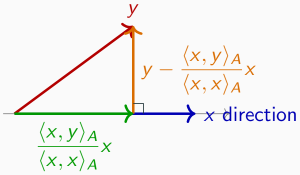
If $\langle z, z\rangle_A = 0$ for every z, then $A = 0$ and Cauchy is trivial.

So the proof is really Case 1 plus Case 2.

Why Build This Vector w?

In the A-geometry, the component of y along x is

$$\frac{\langle x, y\rangle_A}{\langle x, x\rangle_A} x.$$



The orange segment is A-perpendicular to x:

$$\left\langle x,\ y - \frac{\langle x, y\rangle_A}{\langle x, x\rangle_A} x \right\rangle_A = 0.$$

The proof uses this perpendicular segment, then clears the denominator.

From the Segment to w

The perpendicular residual is

$$u = y - \frac{\langle x, y \rangle_A}{\langle x, x \rangle_A} x.$$

To avoid fractions, multiply by $\langle x, x \rangle_A$:

$$\boxed{w = \langle x, x \rangle_A u = \langle x, x \rangle_A y - \langle x, y \rangle_A x.}$$

This scaling does not change the key geometric fact:

$$\langle x, w \rangle_A = 0.$$

w is the perpendicular part of y, scaled for clean algebra.

Case 1: Positive Length

Assume the first case:

$$\langle x, x\rangle_A > 0.$$

Build a vector designed to cancel the x-component of y:

$$w = \langle x, x\rangle_A \, y - \langle x, y\rangle_A \, x.$$

Since A is positive semidefinite:

$$0 \leq \langle w, w\rangle_A.$$

Now expand this one inequality.

Expand $\langle w, w \rangle_A$

Let

$$\alpha = \langle x, x \rangle_A, \qquad \beta = \langle x, y \rangle_A.$$

Then:

$$w = \alpha y - \beta x.$$

So:

$$0 \leq \langle w, w \rangle_A = \langle \alpha y - \beta x, \alpha y - \beta x \rangle_A.$$

Using bilinearity:

$$0 \leq \alpha^2 \langle y, y \rangle_A - 2\alpha\beta \langle x, y \rangle_A + \beta^2 \langle x, x \rangle_A.$$

Finish Cauchy

Since

$$\beta = \langle x, y \rangle_A, \qquad \alpha = \langle x, x \rangle_A,$$

the expansion becomes:

$$0 \leq \alpha^2 \langle y, y \rangle_A - 2\alpha\beta^2 + \beta^2\alpha.$$

Combine the last two terms:

$$0 \leq \alpha^2 \langle y, y \rangle_A - \alpha\beta^2.$$

Finish Cauchy (continued)

We have:

$$0 \leq \alpha^2 \langle y, y \rangle_A - \alpha\beta^2.$$

Divide by $\alpha = \langle x, x \rangle_A > 0$:

$$0 \leq \langle x, x \rangle_A \langle y, y \rangle_A - \langle x, y \rangle_A^2.$$

Therefore:

$$\boxed{\langle x, y \rangle_A^2 \leq \langle x, x \rangle_A \langle y, y \rangle_A.}$$

Case 2: Zero Length

What if $\langle x, x\rangle_A = 0$?

For any real number t:

$$0 \leq \langle x + ty, x + ty\rangle_A.$$

Expanding:

$$0 \leq 2t\langle x, y\rangle_A + t^2\langle y, y\rangle_A.$$

This quadratic cannot be nonnegative for all t unless:

$$\boxed{\langle x, y\rangle_A = 0.}$$

Worked Cross-Filling Tests

Test 1: Zero Diagonal Obstruction

Consider

$$\underbrace{\begin{pmatrix} 1 & 2 & -3 \\ 2 & 0 & 2 \\ -3 & 2 & 5 \end{pmatrix}}_{A}$$

If $A \succeq 0$ and $a_{22} = 0$, the zero diagonal lemma forces the whole second row and column to be zero.

But here:

$$a_{21} = 2, \qquad a_{23} = 2.$$

Therefore A is not positive semidefinite.

Test 2: Negative Remainder Obstruction

Start with

$$\underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 5 & 4 \\ 2 & 4 & 9 \end{pmatrix}}_{B}.$$

Cross-fill from the first diagonal pivot and write the decomposition:

$$\underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 5 & 4 \\ 2 & 4 & 9 \end{pmatrix}}_{B} = \underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 9 & 6 \\ 2 & 6 & 4 \end{pmatrix}}_{P_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & -4 & -2 \\ 0 & -2 & 5 \end{pmatrix}}_{R}.$$

A PSD remainder cannot have a negative diagonal entry.

Test 3: Positive Cross-Filling

Now consider

$$\underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 10 & 7 \\ 2 & 7 & 9 \end{pmatrix}}_{A}.$$

First diagonal cross-fill:

$$\underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 10 & 7 \\ 2 & 7 & 9 \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 9 & 6 \\ 2 & 6 & 4 \end{pmatrix}}_{P_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 5 \end{pmatrix}}_{R_1}.$$

Test 3: Continue

The remaining lower-right block is:

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 5 \end{pmatrix}}_{R_1} = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}}_{P_2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{pmatrix}}_{R_2}.$$

Final pivot:

$$R_2 = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{pmatrix}}_{P_3}.$$

Test 3: Entrywise Sum of Positive Pieces

Expand every rank-one piece as an entrywise matrix:

$$\underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 10 & 7 \\ 2 & 7 & 9 \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 9 & 6 \\ 2 & 6 & 4 \end{pmatrix}}_{P_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}}_{P_2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{pmatrix}}_{P_3}.$$

This is the standard cross-filling output: a sum of filled rank-one matrices.

Test 3: Then Factor Each Piece

Each piece is a column times its transpose:

$$\underbrace{\begin{pmatrix}1\\3\\2\end{pmatrix}\begin{pmatrix}1 & 3 & 2\end{pmatrix}}_{P_1=m_1m_1^T}+\underbrace{\begin{pmatrix}0\\1\\1\end{pmatrix}\begin{pmatrix}0 & 1 & 1\end{pmatrix}}_{P_2=m_2m_2^T}+\underbrace{\begin{pmatrix}0\\0\\2\end{pmatrix}\begin{pmatrix}0 & 0 & 2\end{pmatrix}}_{P_3=m_3m_3^T}.$$

The cross-filled column is always:

$$\boxed{m=\frac{\text{pivot column}}{\sqrt{\text{pivot}}}}\qquad m_1=\frac{1}{\sqrt{1}}\begin{pmatrix}1\\3\\2\end{pmatrix},\quad m_2=\frac{1}{\sqrt{1}}\begin{pmatrix}0\\1\\1\end{pmatrix},\quad m_3=\frac{1}{\sqrt{4}}\begin{pmatrix}0\\0\\4\end{pmatrix}=\begin{pmatrix}0\\0\\2\end{pmatrix}.$$

Test 3: Stack the Columns

$$\underbrace{M = \begin{pmatrix} | & | & | \\ m_1 & m_2 & m_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 2 \end{pmatrix}}_{\text{stack the cross-filled columns}} \quad \Longrightarrow \quad \boxed{A = MM^T.}$$

$$x^T A x = \|M^T x\|^2 = (x_1 + 3x_2 + 2x_3)^2 + (x_2 + x_3)^2 + (2x_3)^2.$$

Cross-filling is both a test and a constructive factorization.

Cross-Filling Criterion

For a real symmetric matrix:

Cross-filling behavior	Conclusion
stuck: zero diagonal, nonzero row/column	indefinite
nonzero pivots have both signs	indefinite
all pivots nonnegative; cross-filling has no difficulties	positive semidefinite
all pivots nonpositive; cross-filling has no difficulties	negative semidefinite
all pivots strictly positive/negative	positive/negative definite

The test records pivots and remainders, not just determinant products.

This is the computational test from diagonal cross-filling.

Connection to Determinants

What the Literature Says

Many textbooks do not say "cross-filling."

They usually say:

$$\boxed{\begin{array}{c}\text{positive definite}\\ \Longleftrightarrow \\ \text{certain principal determinants are positive}\end{array}}$$

For example, in the no-row-swap diagonal order:

$$\det A_1 > 0, \qquad \det A_2 > 0, \qquad \cdots, \qquad \det A_n > 0.$$

This is Sylvester's determinant criterion.

Principal Determinants

For a 3×3 matrix, leading principal determinants are:

$$\underbrace{\begin{pmatrix} a_{11} & * & * \\ * & * & * \\ * & * & * \end{pmatrix}}_{A_1} \quad \underbrace{\begin{pmatrix} a_{11} & a_{12} & * \\ a_{21} & a_{22} & * \\ * & * & * \end{pmatrix}}_{A_2} \quad \underbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}}_{A_3 = A}.$$

The literature checks:

$$\det A_1, \qquad \det A_2, \qquad \det A_3.$$

Our Dictionary: Determinant = Product of Pivots

Diagonal cross-filling produces pivots:

$$d_1,\ d_2,\ \ldots,\ d_n.$$

From our determinant lecture:

$$\boxed{\det A_k = d_1 d_2 \cdots d_k.}$$

So the next pivot is the ratio:

$$\boxed{d_k = \frac{\det A_k}{\det A_{k-1}}.}$$

Principal determinants are just cumulative products of cross-filling pivots.

Positive Definite: Same Test, Different Language

Cross-filling says:

$$d_1 > 0, \qquad d_2 > 0, \qquad \cdots, \qquad d_n > 0.$$

Determinants say:

$$\det A_1 > 0, \qquad \det A_2 > 0, \qquad \cdots, \qquad \det A_n > 0.$$

They are equivalent because:

$$\det A_k = d_1 d_2 \cdots d_k.$$

Same information: determinant criterion hides the pivots inside products.

Why This Breaks for Semidefinite

Leading determinants only see products:

$$d_1 = 0 \quad \Longrightarrow \quad \det A_1 = \det A_2 = \cdots = 0$$

even if some hidden direction is negative.

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}}_{B}$$

Here B is not PSD, but:

$$\det B_1 = 0, \qquad \det B_2 = 0, \qquad \det B_3 = 0.$$

Leading determinant tests cannot certify semidefinite matrices.

Semidefinite Example: Zero in the Middle

Consider

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Leading determinants say only:

$$\det A_1 = 2, \qquad \det A_2 = 0, \qquad \det A_3 = 0.$$

Cross-filling sees the structure:

$$\boxed{2 > 0} \quad \longrightarrow \quad \boxed{0 \text{ with zero row/column}} \quad \longrightarrow \quad \boxed{3 > 0}.$$

One zero pivot is harmless only because its whole row/column is zero.

Indefinite Example: Determinants Hide the Negative Pivot

Now change only the last direction:

$$B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -3 \end{pmatrix}.$$

The leading determinant data still looks inconclusive:

$$\det B_1 = 2, \qquad \det B_2 = 0, \qquad \det B_3 = 0.$$

But cross-filling does not stop at the middle zero:

$$\boxed{2 > 0} \quad \longrightarrow \quad \boxed{0 \text{ with zero row/column}} \quad \longrightarrow \quad \boxed{-3 < 0}.$$

The zero is skipped structurally; the negative pivot is exposed.

Nontrivial Semidefinite Example

Now the zero pivot is hidden:

$$A = \begin{pmatrix} 2 & 2 & 4 \\ 2 & 2 & 4 \\ 4 & 4 & 11 \end{pmatrix}.$$

First cross-fill from the pivot 2:

$$A - \underbrace{\begin{pmatrix} 2 & 2 & 4 \\ 2 & 2 & 4 \\ 4 & 4 & 8 \end{pmatrix}}_{P_1} = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}}_{R_1}.$$

So the pivots are:

$$\boxed{2 > 0} \quad \longrightarrow \quad \boxed{0 \text{ with zero row/column}} \quad \longrightarrow \quad \boxed{3 > 0}.$$

The semidefinite zero appears only after subtracting the first filled block.

Nontrivial Indefinite Example

Use almost the same matrix:

$$B = \begin{pmatrix} 2 & 2 & 4 \\ 2 & 2 & 4 \\ 4 & 4 & 5 \end{pmatrix}.$$

First cross-fill from the pivot 2:

$$B - \underbrace{\begin{pmatrix} 2 & 2 & 4 \\ 2 & 2 & 4 \\ 4 & 4 & 8 \end{pmatrix}}_{P_1} = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -3 \end{pmatrix}}_{R_1}.$$

Thus:

$$\boxed{2 > 0} \longrightarrow \boxed{0 \text{ with zero row/column}} \longrightarrow \boxed{-3 < 0}.$$

The determinant products see zero; cross-filling exposes the negative direction.

Semidefinite: Why Literature Becomes Heavier

For positive semidefinite matrices, zero pivots can appear.

Then leading determinants alone are not enough. Textbooks usually say:

$$\boxed{\text{all principal minors must be nonnegative.}}$$

That means many determinants, even in 3×3:

$$\begin{gathered} \det A_{\{1\}}, \quad \det A_{\{2\}}, \quad \det A_{\{3\}}, \\ \det A_{\{1,2\}}, \quad \det A_{\{1,3\}}, \quad \det A_{\{2,3\}}, \\ \det A_{\{1,2,3\}}. \end{gathered}$$

Our zero-row rule replaces this determinant explosion with structure.

Complexity: Determinants vs. Pivots

For an $n \times n$ matrix, all principal minors means one determinant for every nonempty subset:

$$\boxed{2^n - 1 \text{ determinants.}}$$

Cross-filling checks the diagonal behavior directly:

$$\boxed{\text{pivots } d_k \quad + \quad \text{zero row/column rule} \quad + \quad \text{remainders.}}$$

Literature PSD test	**Cross-filling PSD test**
many principal determinants	pivots and remainders
products hide zero pivots	zero pivots are inspected
sign answer only	structural obstruction

Why Cross-Filling Is Stronger

Determinants answer only one question:

$$\det(\text{submatrix}) \quad \text{positive, zero, or negative?}$$

Cross-filling gives more data:

Cross-filling data	**Meaning**
pivot d_k	determinant ratio
rank-one piece P_k	squared-length component
remainder R_k	next matrix to check
zero row/column	semidefinite degeneracy
negative diagonal in R_k	obstruction

Computing determinants is essentially cross-filling with less memory.

Applications

Application 1: Square Root of a PSD Matrix

$$\underbrace{A \succeq 0}_{\text{positive semidefinite}} \quad \Longrightarrow \quad \underbrace{A = \Omega \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \Omega^T}_{\substack{\text{orthogonal diagonalization} \\ \lambda_i \geq 0}} .$$

$$\underbrace{A^{1/2} := \Omega \begin{pmatrix} \sqrt{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & \sqrt{\lambda_n} \end{pmatrix} \Omega^T}_{\text{define one symmetric square root}} .$$

$$\underbrace{A^{1/2} A^{1/2}}_{\text{multiply in the same eigenbasis}} = \Omega \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \Omega^T = A.$$

Matrix square roots are not unique; this defines the symmetric PSD square root.

Application 1: Example

$$\underbrace{A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}}_{\text{PSD}} = \underbrace{\Omega}_{\text{orthogonal eigenbasis}} \underbrace{\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}}_{\Lambda} \underbrace{\Omega^T}_{\text{eigen-coordinate rows}} .$$

$$\underbrace{A^{1/2} := \Omega \begin{pmatrix} \sqrt{4} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \Omega^T}_{\text{replace squared axis coefficients by axis coefficients}} = \underbrace{\Omega \begin{pmatrix} 2 & 0 \\ 0 & \sqrt{2} \end{pmatrix} \Omega^T}_{\text{symmetric square root}} .$$

PSD matrices allow a real symmetric square root.

Application 2: Comparing Two Inner Products

$$\underbrace{A = \Omega \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \Omega^T}_{\lambda_{\min} \leq \lambda_i \leq \lambda_{\max}} \qquad \underbrace{y = \Omega^T x}_{\text{orthogonal change of coordinates}} \qquad .$$

$$\underbrace{x^T A x}_{\langle x, x \rangle_A} = y^T \underbrace{\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}}_{\lambda_1 y_1^2 + \cdots + \lambda_n y_n^2} y \, .$$

$$\underbrace{\lambda_{\min}(y_1^2 + \cdots + y_n^2)}_{\lambda_{\min} \|x\|^2} \leq \underbrace{x^T A x}_{\text{new squared length}} \leq \underbrace{\lambda_{\max}(y_1^2 + \cdots + y_n^2)}_{\lambda_{\max} \|x\|^2} \, .$$

Eigenvalue Bound for the Ratio

$$\boxed{\lambda_{\min} \le \underbrace{\frac{x^T A x}{x^T x}}_{\text{ratio of new squared length to old squared length}} \le \lambda_{\max} \qquad (x \neq 0).}$$

Theorem (Eigenvalue Comparison)

If A is real symmetric with smallest eigenvalue $\lambda_{\min}$ and largest eigenvalue $\lambda_{\max}$, then every nonzero vector x satisfies

$$\boxed{\lambda_{\min} \le \frac{x^T A x}{x^T x} \le \lambda_{\max}.}$$

Eigenvalues measure the strongest and weakest stretching of the inner product.

Final Summary

Complete answer:

$$A = M^T M \quad \iff \quad x^T A x \geq 0 \quad \text{for all } x.$$

Test	PSD criterion
quadratic form	$x^T A x \geq 0$ for all x
eigenvalues	every $\lambda_i \geq 0$
factorization	$A = M^T M$
cross-filling	positive pivots + zero-row rule

Positive definite means the same story with strict positivity.